Dispersed Phase - Model

- Number density function
- PBE
- Breakage (source & sink)
- Aggregation (aerosol dynamics)
  - Ex :- cloud formation
    mist , droplets
    sintering

Aggregation

- Aggregation frequency
&
- Pair Correlation (distribution) function

$f_1(x) \rightarrow$ Number density function

$f_2(x; x') \rightarrow$ distinct pair density function which are aggregating.
(two particles)
- Binary (✓)

f_3, f_4, ... Triplet , Quadruplet (✗)

particles which are aggregating

$x , x' \longrightarrow f_1(x) , f_2(x')$

new state, $\bar{x} \longrightarrow f_1(\bar{x})$

Number density function

$a(x, r; x', r'; t) \rightarrow$ aggregating frequency

PBE : RHS $\rightarrow$ h^{+} + h^{-1}

(new state) (source) sink (old state)

$$h^{+} = \int_{\Omega_R} dV_x \int_{\Omega_R} \frac{1}{\delta}\, dV_r\; a(\tilde{x}, \tilde{r}; x', r', t)\, f_2(\tilde{x}, \tilde{r} : x', r', t) \boxed{\frac{\partial(\tilde{x}, \tilde{r})}{\partial(x, r)}}$$

$\uparrow$ $h^{+}(x, r, t)$

$\downarrow$ Duplicating factor

$\downarrow$ relative factor relating new state which will be the kind of old state for next

$$a(\tilde{x}, \tilde{r}; x', r', t) = a(\tilde{x}, \tilde{r}; x, r, t)$$

$$\frac{\partial(\tilde{x}, \tilde{r})}{\partial(x, r)} = \begin{bmatrix} \frac{\partial \tilde{x}_1}{\partial x_1} & \frac{\partial \tilde{x}_2}{\partial x_2} & \cdots & \frac{\partial \tilde{x}_n}{\partial x_n} & \frac{\partial \tilde{x}_1}{\partial r_1} & \frac{\partial \tilde{x}_1}{\partial r_2} & \frac{\partial \tilde{x}_1}{\partial r_3} \\ \vdots \\ \frac{\partial \tilde{x}_n}{\partial x_1} \\ \frac{\partial \tilde{r}_1}{\partial x_1} \\ \vdots \\ \frac{\partial \tilde{r}_3}{\partial x_1} \end{bmatrix}$$

$$h^{+}(\tilde{x}, \tilde{r}, t) = \int dV_x \int dV_r\; a(x, r; \tilde{x}, \tilde{r}, t)\, f_2(x, r; \tilde{x}, \tilde{r}, t)$$

Pair function: $f_2(x, r; \tilde{x}, \tilde{r}, t) = f_1(x, r, t)\, f_1(x', r', t)$

Aerosol dynamics

Range of particle size, 'nm $\rightarrow$ mm'

6 ords of magn variation

Divided into two parts
- $\rightarrow$ Discrete (small size Range) $\boxed{x_i = i\, x_0}$ $\rightarrow$ $x_0 < x < x_n$ $\hookrightarrow (1 < i < n)$
- $\rightarrow$ Continuous $\rightarrow$ $x_n < x < \infty$

$$N(t) = \sum_{i=1}^{n} f_{1,i}(t) + \int_{x_n}^{\infty} f_1(x, t)\, dx$$

Note

1. There could be jump from discrete range to continuous range
2. Aggregating frequency: a_{ij} (discrete ranges: x_i & x_j)

$a(x, x')$ [continuous range]

$a_i(x)$ [one particle from discrete, one particle in continuous]

3. There is no growth. 4. Lumped system

S

PBE: $\dfrac{\partial f_{1,i}(t)}{\partial t} = h_i^+ - h_i^-(t)$ [Discrete Range]

$\dfrac{\partial f_1(x,t)}{\partial t} = h^+(x,t) - h^-(x,t)$ [continuous Range]

Discrete Range:

$i=1$

Kronecker ~~dirac~~ delta

$$h_1^+(t) = \sum_{j=2}^{n} e(x_j) f_{1,j} (1 + \delta_{j,2}) + \int_{x_n} e(x) f_1(x,t)\,dx$$

→ Smallest particle can only happen due to evaporation, not because of aggregation

$1 < i < n \to \delta$

$$\delta_{n,m} = 1 \; ; n = m$$
$$= 0 \; ; m \neq m$$

$$2 < i < n \rightarrow h_i^+(t) = \frac{1}{2}\sum_{j=1}^{i-1} a_{i-j,j} \underbrace{f_{1,i-j} f_{1,j}}_{f_2(i-j,j)} + \delta_n \int_{x_n}^{x_n + x_1} e(x) f_1(x,t)\,dx + \underbrace{e(x_{i+1}) f_{1,i+1}}_{\text{eva from discrete}}$$

considering binary

aggrega

condn → evapora

$$h_i^-(t) = f_{1,i}\left\{ \sum_{j=1}^{n} a_{i,j} f_{1,j} + \int_{x_n}^{\infty} a_i(x) f_1(x,t)\,dx \right\} + e(x_i) f_{1,i}$$

Continuous range: $x_n \leq x < \infty$

$$\frac{\partial f_1(x,t)}{\partial t} = h^+(x,t) - h^-(x,t)$$

✓ For Discrete to $\overset{joint}{}$ go Continuous range —

$$x_i + x_j = x_{i+j} \quad \text{where } i+j \not> n$$

↓ what should be the mass

$$x_{i+j} < x < x_{i+j} + x_0 \rightarrow \text{Size range}$$

we distribute the mass which formed ↘ $(1/x_0)$
↳ to the distribution in continues.

$$h^+(x,t) = \underbrace{\left[\frac{1}{2x_0}\sum_{j=\left[\frac{x}{x_0}\right]-n}^{n} a_{[x/x_0]-j,\,j}\, f_{1,[x/x_0]-j}\, f_{1,j}\right]\left(H(x - x_{[x/x_0]}) - H(x - x_{[x/x_0]+1})\right) \times \left(1 - H(x - x_{2n})\right)}_{\text{Discrete}}$$

(source)

$$+\left[H(x - x_{n+1})\sum_{j=1}^{n} a_j(x - x_j)\, f_{1,j}\, f_1(x - x_j, t)\right] \rightarrow \text{contin + discr}$$

+

$$\boxed{H(x - x_{i+j}) - H(x - x_{i+j+1}) = \begin{cases} 1 \; ; & x_{i+j} < x < x_{i+j} + x_0 \\ 0 \end{cases}}$$

↳ Heavy side function

$$+\frac{1}{2}\int_{x_n}^{\infty} a(x', x - x')\, f_1(x', t)\, f_1(x - x', t)\, dx \Big] \text{ cont + cont}$$

$$+\frac{1}{2}\, e(x + x_0)\, f_1(x + x_0, t) \Big] \text{ evaporation.}$$

$$h^-(x,t) = f_1(x,t)\left[\sum_{j=1}^{n} a_j(x)\, f_{1,j} + \int_{x_n}^{\infty} a(x, x')\, f_1(x', t)\, dx'\right]$$

(Sink)

$$+\, e(x)\, f_1(x,t)$$

$$f_1(0) = u_i \quad \text{when } i = 0, 1, 2 - - \infty$$

$$f_1(x, 0) = u(x) \quad \forall\, x \geq x_n$$

Typical Aggregation Problem

$$\frac{\partial f_1(x,t)}{\partial t} = -f_1(x,t)\int_0^{\infty} f_1(x',t)\, a_0\, dx' + \frac{1}{2}\int_0^{x^*} a_0 \underbrace{f_2(x',t)}_{f_1(x',t)\, f_1(x-x',t)} dx'$$

Initial condition, $f_1(x,0) = N_0$

$$\boxed{\tau = a_0 N_0 t} \rightarrow \text{Non-dimensional}$$

$$\frac{\partial f_1(x,t)}{\partial \tau} = -\frac{f_1(x,t)}{N_0}\int_0^{\infty} f_1(x',t)\, dx' + \frac{1}{2}\int_0^{x^*} \frac{f_1(x-x',t)}{N_0} f_1(x',t)\, dx'$$

$$f = f_1/N_0$$

$$\frac{\partial f(x,t)}{\partial \tau} = -f(x,t)\int_0^{\infty} f(x',t)\, dx' + \frac{1}{2}\int_0^{x^*} f(x-x',t)\, f(x',t)\, dx'$$

$$\boxed{\eta(\tau) = \frac{N(t)}{N_0} = \frac{1}{N_0}\int_0^{\infty} f_1(x,t)\, dx = \int_0^{\infty} f(x,t)\, dx}$$

If we integrate both side —

LHS

$$\int_0^{\infty} \frac{\partial f(x,t)}{\partial \tau}\, dx = \frac{d}{d\tau}\underbrace{\int_0^{\infty} f(x,t)\, dx}_{\eta} = \frac{d\eta}{d\tau}$$

RHS

$$-\int_0^{\infty} f(x,t)\, \eta(\tau)\, dx = \eta^2 \qquad \int_0^{x'} + \int_{x'}^{\infty} \rightarrow \text{In this case } \underset{\text{const}}{f(u,t)}$$

$$\therefore \frac{d\eta}{d\tau} = -\eta^2 + \frac{1}{2}\int_0^{\infty} dx \int_0^{x^*} f(\underbrace{x-x'}_{u},t)\, f(x',t)\, dx'$$

Let $u = x - x'$ | $x \to x' \to u = 0$

$x \to \infty \to u \to \infty$

$$\frac{d\eta}{d\tau} = -\eta^2 + \frac{1}{2}\underbrace{\int_0^\infty du\, f(u,t)}_{\eta}\underbrace{\int_0^\infty dx'\, f(x',t)}_{\eta}$$

$$\frac{d\eta}{d\tau} = -\eta^2 + \frac{1}{2}\eta^2 = -\frac{1}{2}\eta^2$$

$$\int_{u=0}^\infty du = \int_{x'}^\infty dx'$$

$$-\frac{d\eta}{\eta^2} = \frac{d\tau}{2}$$

$$\frac{1}{\eta} - 1 = \frac{1}{2}\tau$$

$$\boxed{\eta(0) = 1}$$

$$\frac{1}{\eta} = \frac{\tau+2}{2}$$

$$\boxed{\eta = \frac{2}{\tau+2}}$$

$$\eta = \frac{1}{N_0}\int_0^\infty f_1(x,t)\, dx'$$

Laplace Transform

$$\frac{\partial f}{\partial t} = -f(x,t)\int_0^\infty f(x',t)\,dx' + \frac{1}{2}\int_0^{x} f(x-x',t)\, f(x',t)\, dx'$$

Laplace

$$\mathcal{L}(f(x,t)) = \int_0^\infty e^{-sx} f(x,t)\, dx \equiv \bar{f}(s,t)$$

↳ x-domain

$$\frac{\partial \bar{f}(s,t)}{\partial \tau} = -\eta(\tau)\,\bar{f}(s,t) + \frac{1}{2}\int_0^\infty e^{-sx}\,dx \int_0^x f(x-x',t)\, f(x',t)\,dx'$$

$$+\frac{1}{2}\int_{x'=0}^{\infty} dx' \int_0^{\infty} du\, e^{-s(x'+u)} f(u,t)\, f(x'\ldots$$

$$\parallel$$

$$\frac{1}{2}\left(\bar{f}(s,t)\right)^2$$

$x - x' = u$

$x = u + x'$

$$\boxed{\frac{\partial \bar{f}}{\partial \tau} = -\eta \bar{f} + \frac{1}{2}\bar{f}^2}$$

$$\frac{2}{2+\tau}$$

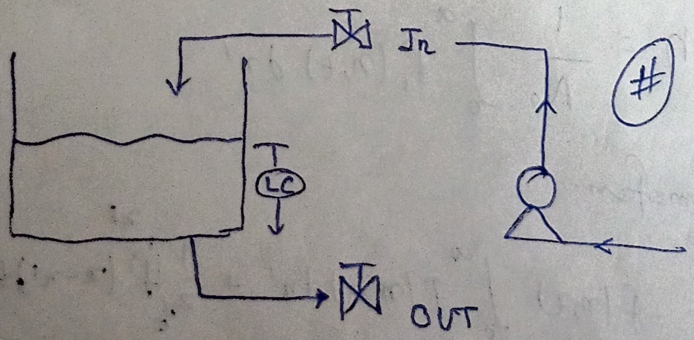
$$\bar{f} = \frac{1}{(2+\tau)^2} \cdot \frac{1}{c_1 + \frac{1}{2(\tau+2)}}$$

inv

$f_1(x,t)$ $\quad c_1 \equiv$ eval from $\bar{f}(s,0)$

$$N_0 = \int_0^{\infty} f_1(x,0)\, dx \quad ; \quad f_1(x,0) = g(x) \rightarrow \bar{g}(s)$$

Programmable Logic Controller (PLC)



States or Modes of PLC

→ Steady
→ Temporal
→ Hybrid

Logic state

1 → ON
0 → OFF

$\rightarrow$ (4 - 20 mV) mA

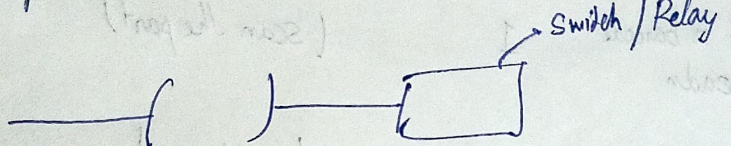
1 → ON / TRUE / contact closure / open / energise etc

0 → OFF / False / contact open / closed / de-energise etc.

Programming Representation

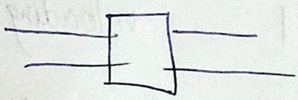
Contact

Coil (output state)



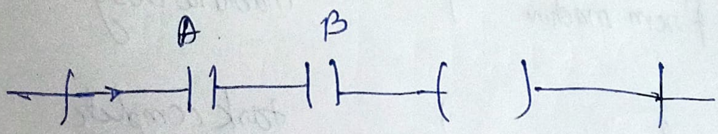
energised when power / current flows.

Boxes

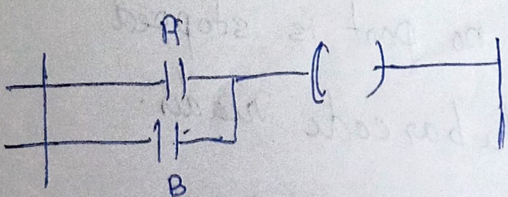


Contain function which are executed when Power / current flow through them.

AND



OR



↓

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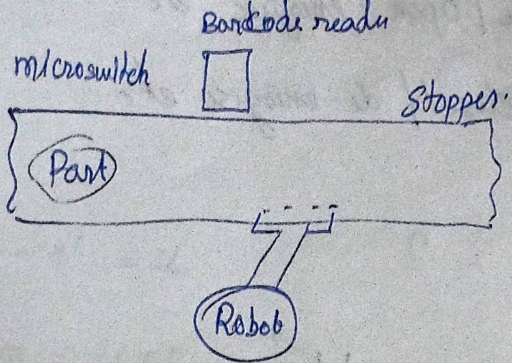


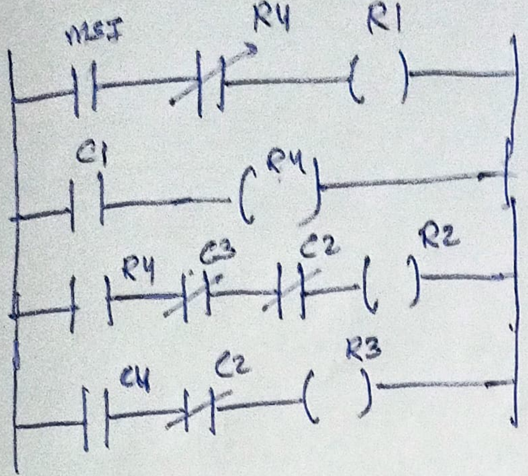
id	desc	State	
MSI	microswitch	1	(part arrive)
R1	output to barcode reader	1	(scan the part)
C1	input from barcode reader	1	(right part)
R2	output robot	1	loading cycle
R3	output "	1	unloading cycle
C2	input from robot	1	robot busy
R4	output to stopper	1	stopper up
C3	input from machine	1	machine busy
C4	"	1	task complete

Rung 1: part arrives & no part is stopped trigger the barcode reader.

Rung 2: If part arrives, actuate the stopper

Rung 3: stopper is up, machine is not busy, Robot is not busy, load onto the machine

Rung 4: Task complete, robot is not busy, unloaded the machine.



Ladder Diagram.